Ders 2

Enerji: $E_x = \lim_{T\to\infty} \int_{-T/2}^{T/2} |x(t)|^2 dt = \int_{-\infty}^{\infty} |x(t)|^2 dt$

$E_x < \infty \Rightarrow$ Enerji işaretidir

Güç: $P_x = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt$

$0 < P_x < \infty \Rightarrow$ Güç işaretidir

$x(t) = \begin{cases} A\cos(2\pi f_0 t + \theta) \\ A\sin(2\pi f_0 t + \theta) \end{cases} \longrightarrow P_x = \frac{A^2}{2}$

$T_0 = \frac{1}{f_0}$ periyottur

Periyodik $\Rightarrow P_x = \frac{1}{T_0} \int_{T_0} |x(t)|^2 dt$

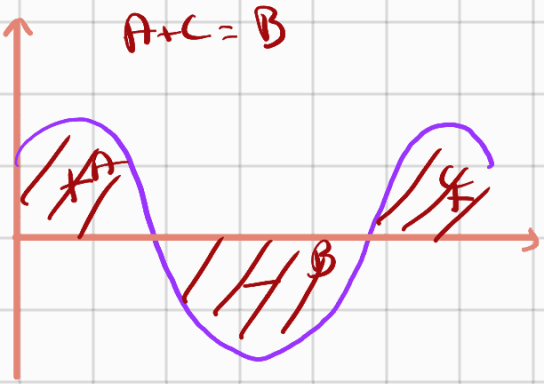
İspat

$$P_x = \frac{1}{T_0} \int_{T_0} A^2 \cos^2(2\pi f_0 + \theta) dt = \frac{A^2}{2T_0} \int_{-T_0/2}^{T_0/2} [1 + \cos(4\pi f_0 + 2\theta)] dt$$

$$= \frac{A^2}{2T_0} \left[T_0 + \int_{-T_0/2}^{T_0/2} \cos(4\pi f_0 t + 2\theta) dt \right]$$

$\int_{-T_0/2}^{T_0/2} \cos(4\pi f_0 t + 2\theta) dt = 0$

* sinüsoidal işaretlerin bir periyot boyunca integrali 0'dır.



$$\Rightarrow \boxed{P_x = \frac{A^2}{2}}$$

Örnek: Lathi'den örnek

$x(t) = c_1 \cdot \cos(2\pi f_1 t + \theta_1) + c_2 \cos(2\pi f_2 t + \theta_2)$, $f_1 \neq f_2$

$P_x = ?$

Bir işaretin periyodunu bileşenlerinin periyodlarının en küçük ortak katlarıyla bulunabilir

$$P_x = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt$$

$$= \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} c_1^2 \cos^2(2\pi f_1 t + \theta_1)\, dt + \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} c_2^2 \cos^2(2\pi f_2 t + \theta_2) dt$$

$c_1^2/2$ $\qquad$ $c_2^2/2$

$$+ \lim_{T\to\infty} \frac{1}{T} 2c_1c_2 \int_{-T/2}^{T/2} \cos(2\pi f_1 t + \theta_1) \cos(2\pi f_2 t + \theta_2)\, dt$$

$$\lim_{T\to\infty} \frac{1}{T} c_1c_2 \int_{-T/2}^{T/2} \left[\cos(2\pi(f_1+f_2)t + \theta_1 + \theta_2) + \cos(2\pi(f_1-f_2)t + \theta_1 - \theta_2)\right] dt$$

$\cos\alpha \cdot \cos\beta = \frac{1}{2}\left[\cos(\alpha+\beta) + \cos(\alpha-\beta)\right]$

$\to 0$ $\qquad$ $\to 0$ $\qquad$ çünkü sinüsoidal

$$\Rightarrow P_x = \begin{cases} \frac{c_1^2}{2} + \frac{c_2^2}{2} & f_1 \neq f_2 \\ \frac{c_1^2}{2} + \frac{c_2^2}{2} + c_1 c_2 \cos(\theta_1 - \theta_2) & f_1 = f_2 \end{cases}$$

burdan bu tercih gelir

Sonuçlar

$x(t) = x_1(t) + x_2(t) \Rightarrow P_x \neq P_{x_1} + P_{x_2}$

→ compact fourier açılımı

$$x(t) = \sum_{n=1}^{\infty} c_n \cos(2\pi f_n t + \theta_n) \Rightarrow P_x = \frac{1}{2} \sum_{n=1}^{\infty} c_n^2 \quad \text{if } f_i \neq f_j$$

(Parseval teoremi)

Örnek

$x(t) = A e^{j2\pi f_0 t}$

$$P_x = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} |x(t)|^2 dt = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} \underbrace{|A e^{j2\pi f_0 t}|^2}_{A} dt$$

$$= \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} |A|^2 dt = |A|^2$$

$z = x + jy = |z| e^{j\alpha}$

$|z| = \sqrt{x^2+y^2}$

$\theta = \arctan \frac{y}{x}$

$|z_1 \cdot z_2| = |z_1||z_2|$

$|A e^{j2\pi f_0 t}| = |A|\,|e^{j2\pi f_0 t}|$

$\cos(\alpha) + j\sin(\alpha)$

Bazı önemli işaretler

Sinüsoidal işaretler

$$x(t) = A\cos(2\pi f_0 t + \theta)$$

- A: Genlik (Amplitude)
- f_0: Frekans ($Hz = \frac{1}{s}$)
- θ: Faz (rad) (phase)

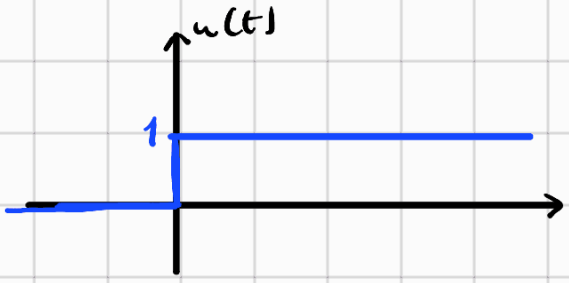
Ağ üç veya enerji işaretin genliği ile orantılıdır

Kompleks üstel işaretler:

$$x(t) = A e^{j(2\pi f_0 t + \theta)}$$

Birim Basamak İşareti:
(Unit step signal)

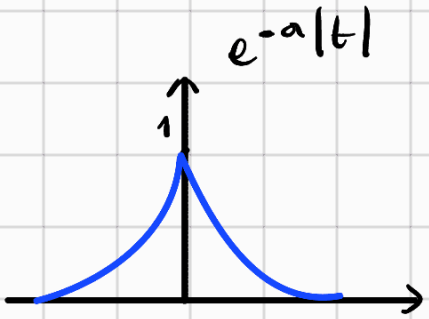
$$u(t) = \begin{cases} 1 & t \geq 0 \\ 0 & t < 0 \end{cases}$$

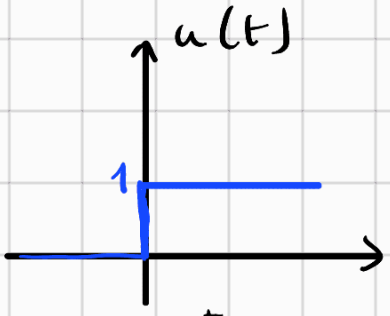


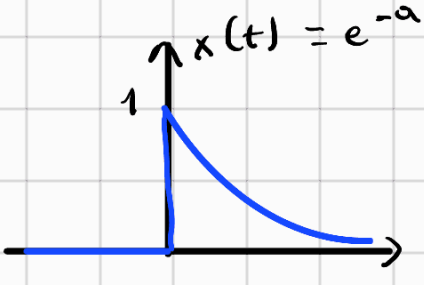
$a > 0 \quad u(at) = u(t)$

Örnek

$$x(t) = e^{-a|t|}\, u(t), \quad a > 0$$



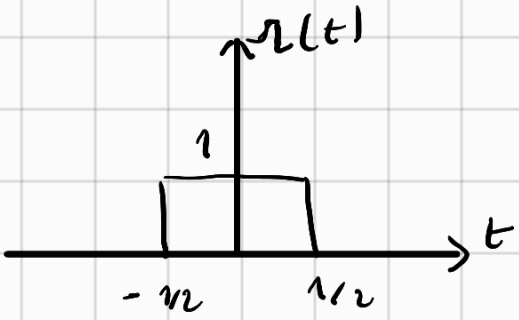




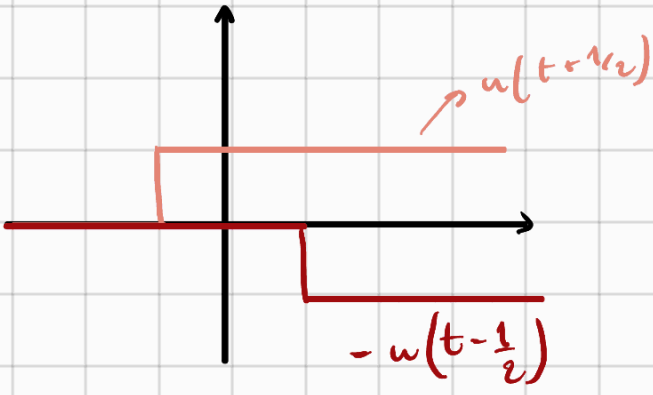
$$u(t) = \begin{cases} e^{-at}, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

Dikdörtgen Darbe işareti
(Rectangular pulse Signal)

$$rect(t) = \Pi(t) = \begin{cases} 1, & -1/2 \le t \le \frac{1}{2} \\ 0, & \text{Diğer} \end{cases}$$

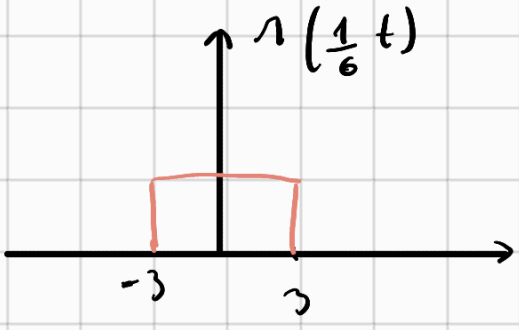


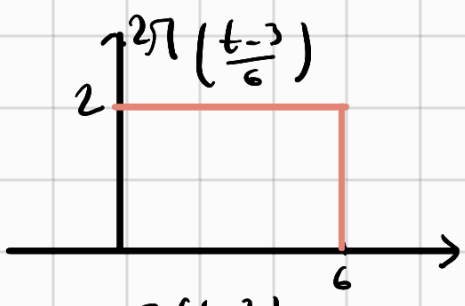
$$\Pi(t) = u\left(t+\frac{1}{2}\right) - u\left(t-\frac{1}{2}\right)$$

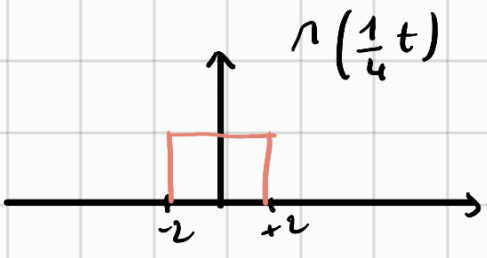


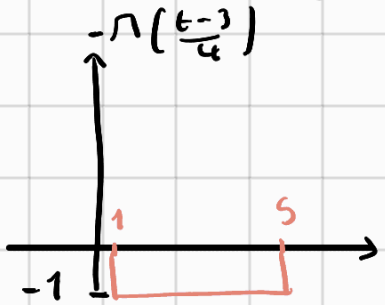
Örnek $x(t) = 2\Pi\left(\frac{t-3}{6}\right) - \Pi\left(\frac{t-3}{4}\right)$ çiziniz

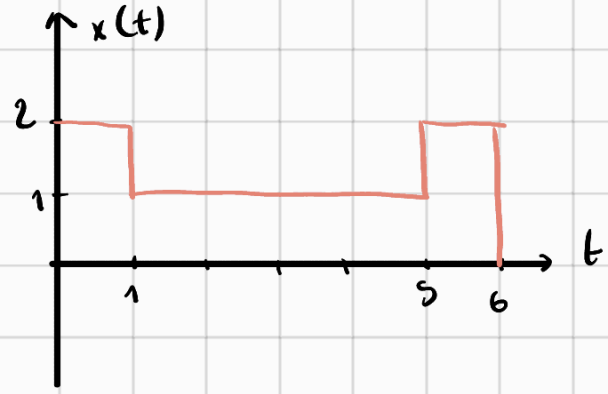
$\frac{1}{6}(t-3)$ $\qquad$ $\frac{1}{4}(t-3)$







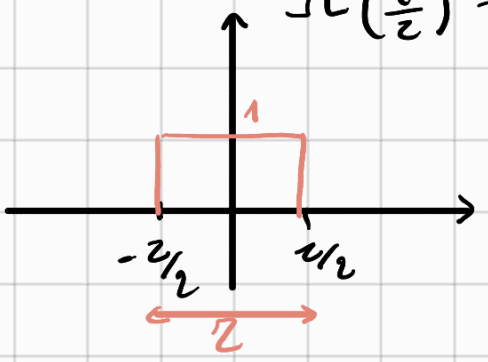




$$\Pi(t) = \begin{cases} 1 & -\frac{1}{2} \le t \le \frac{1}{2} \\ 0 & \text{diğer} \end{cases}$$

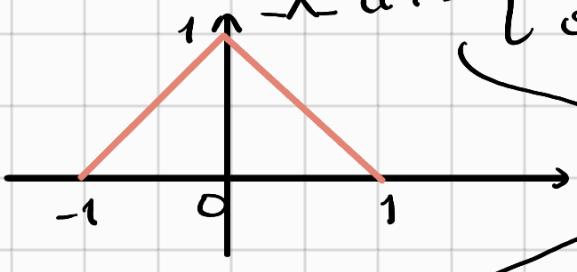


$$\Pi\left(\frac{t}{2}\right) = \begin{cases} 1, & -\frac{2}{2} \le t \le \frac{2}{2} \\ 0, & \text{diğer} \end{cases}$$



Üçgen darbe işareti
(Triangular Pulse Signal)

$$\Lambda(t) = \begin{cases} t+1 & -1 \le t \le 0 \\ -t+1 & 0 \le t \le 1 \\ 0 & \text{Diğer} \end{cases}$$



Doğrunun denklemi

eğim
$y = mt + n$

$t_1, y_1 \rightarrow$

$t_2, y_2 \rightarrow$

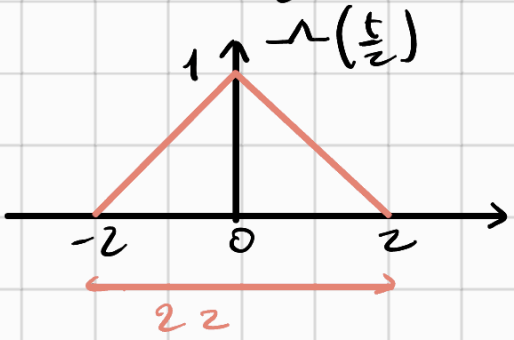
$-1 \le t \le 1$

$$\Lambda(t) = \begin{cases} 1 - |t|, & |t| \le 1 \\ 0, & \text{diğer} \end{cases}$$

$$\Lambda(t) = \Pi(t) \circledast \Pi(t) \longrightarrow \boxed{2.\ \Lambda\left(\frac{t}{2}\right) = \Pi\left(\frac{t}{2}\right) \circledast \Pi\left(\frac{t}{2}\right)}$$

$$x(t) \circledast y(t) = \int_{-\infty}^{+\infty} x(z)\, y(t-z)\,dz = \int_{-\infty}^{+\infty} x(t-z)\, y(z)\,dz$$

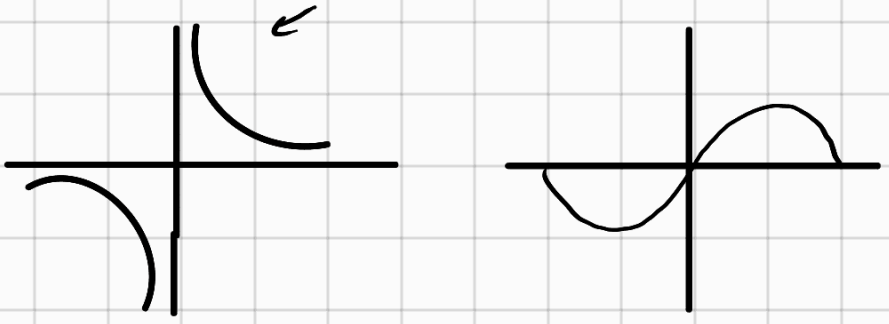
$$\wedge\left(\frac{t}{2}\right) = \begin{cases} 1 - \frac{|t|}{2}, & \frac{|t|}{2} \le 1 \\ 0, & \text{diğer} \end{cases} \quad : |t| \le 2 \quad : -2 \le t \le 2$$

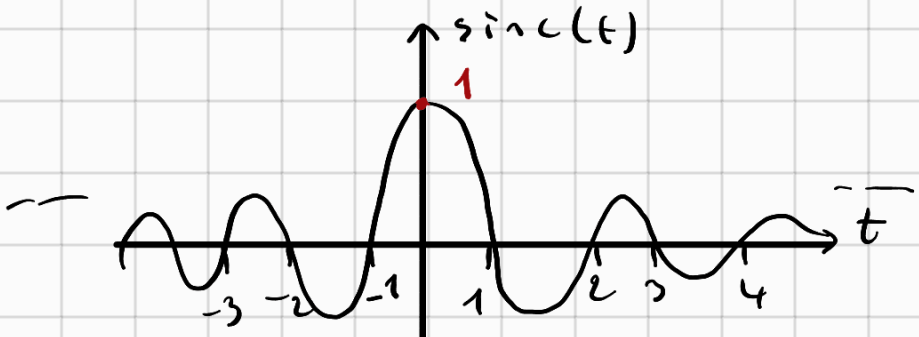


Konvülasyon sonucunda elde edilen işaretin genişliği konvülasyona giren işaretlerinin genişliğinin toplamıdır.

Sinc Darbe işareti
(Sinc Pulse Signal)

$$\text{Sinc}(t) = \frac{\sin(\pi t)}{\pi t}$$





* $(t=k) \in Z-\{0\} \Rightarrow \text{Sinc}(k) = \frac{\sin(k\pi)}{k\pi} = 0$

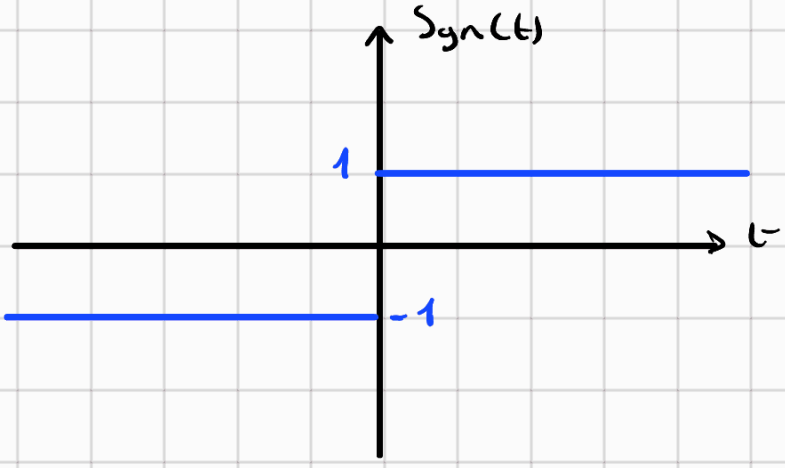
$t=0 \Rightarrow \text{sinc}(0) = \frac{\sin(0)}{0} = \frac{0}{0}$

Hopital: $\lim_{t\to 0} \frac{\pi \cdot \cos(\pi t)}{\pi} = 1$

$\text{sinc}(-t) = \text{Sinc}(t)$

İşaret fonksiyonu
(sign / signum function)

$$Sgn(t) = \begin{cases} 1 & t>0 \\ -1 & t<0 \\ 0 & t=0 \end{cases}$$

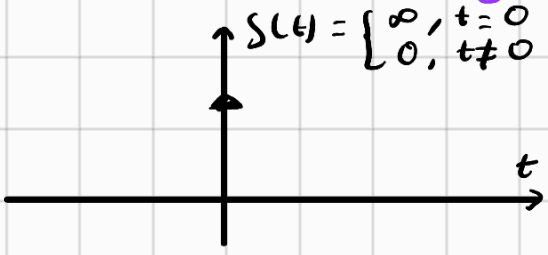


$$Sgn(t) = u(t) - u(-t)$$

$$u(t) = \frac{1}{2}(1 + Sgn(t))$$

Dürtü işareti
(Impulse (Delta) Signal)

$$\delta(t) = \begin{cases} \infty, & t=0 \\ 0, & t\neq 0 \end{cases}$$



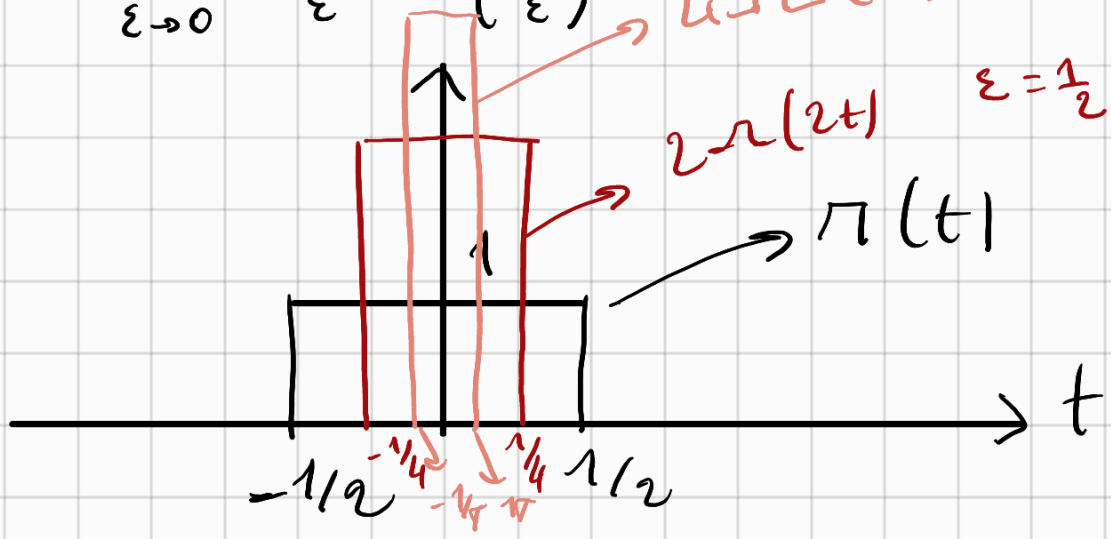
Alan = şiddet = $\int_{-\infty}^{+\infty} \delta(t)\,dt = 1$ //

$$\int_{-\infty}^{+\infty} f(t)\,\delta(t)\,dt = f(0)$$ (sifting property)

f(t) t=0'da sürekliyse

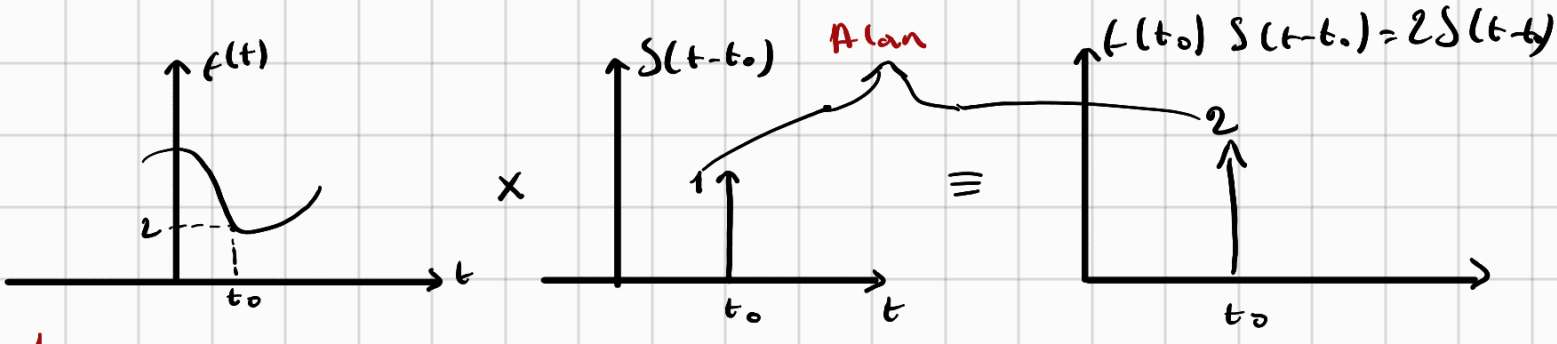
sinc de bu koşulu sağlıyor

$$\delta(t) = \lim_{\varepsilon \to 0} \frac{1}{\varepsilon} \Pi\left(\frac{t}{\varepsilon}\right)$$



Özellikleri : sifting/sampling property

1) $f(t)\,\delta(t-t_0) = f(t_0)\,\delta(t-t_0)$, t_0 = sabit

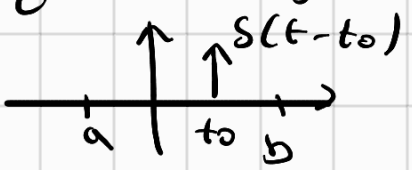


2) $f(t)$ $t = t_0$'da sürekli olmak üzere

$$\int_a^b f(t)\,\delta(t-t_0)\,dt = \begin{cases} f(t_0) & a < t_0 < b \\ 0 & \text{diğer} \end{cases}$$

$f(t) = 1$ →

$$\int_a^b \delta(t-t_0)\,dt = \begin{cases} 1 & 0 < t_0 < b \\ 0 & \text{diğer} \end{cases}$$



$f(t_0)\,\delta(t-t_0)$

$f(t_0) \int_a^b \delta(t-t_0)\,dt$

$1,\ a < t_0 < b$

3) $\forall a \neq 0 \qquad \delta(at) = \frac{1}{|a|}\delta(t)$

$a = -1 \Rightarrow \quad \delta(-t) = \frac{1}{|-1|}\delta(t)$

$\Rightarrow \delta(-t) = \delta(t)$ çift fonksi gonder.

4) $f(t) \circledast \delta(t) = f(t)$

★ $f(t) \circledast \delta(t-t_0) = f(t-t_0)$

$$x(t) \circledast y(t) = \int_{-\infty}^{\infty} x(z)\, y(t-z)\, dz$$

$$f(t) \circledast \delta(t) = \int_{-\infty}^{+\infty} f(z)\, \delta(t-z)\, dz$$

$$= \int_{-\infty}^{+\infty} f(z)\, \delta(z-t)\, dz$$

$$= f(t) \;//$$

$$f(t) \circledast \delta(t-t_0) = \int_{-\infty}^{+\infty} f(z) \underbrace{\delta(t-z-t_0)}_{\delta(z-\underbrace{(t-t_0)}_{\text{sabit}})}\, dz$$

$$= f(t-t_0)$$

5) $$\int_{-\infty}^{t} \delta(z)\, dz = \begin{cases} 1 & , t>0 \\ 0 & , t<0 \end{cases} \equiv u(t)$$

$$\int_{-\infty}^{t} \delta(z)\, dz = u(t) \xrightarrow{\frac{d}{dt}} \frac{d}{dt} u(t) = \delta(t)$$

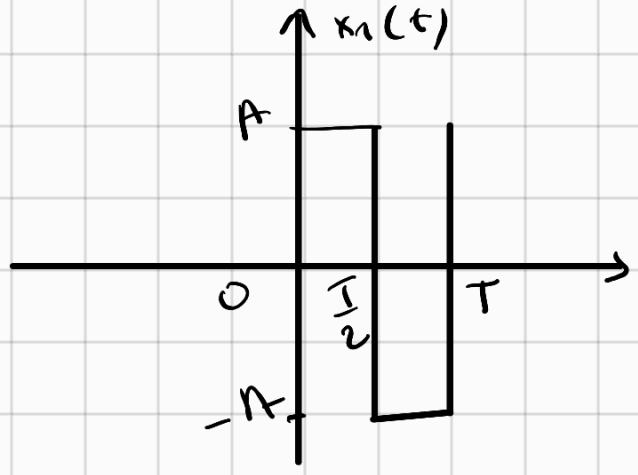
Ortogonal (Dik) / Ortonormal işaretleri:

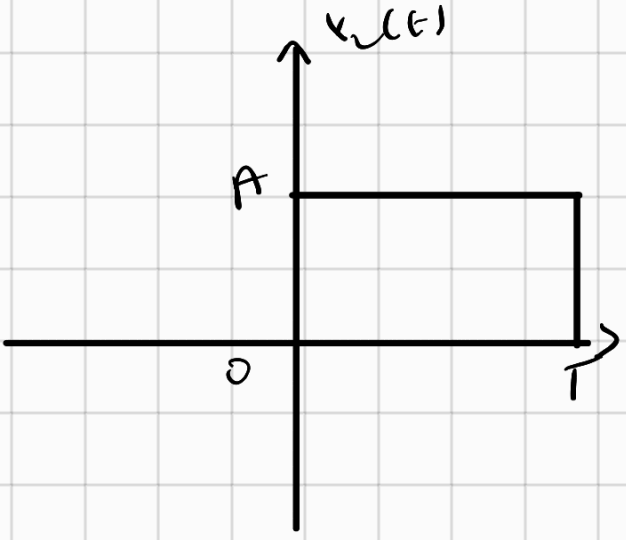
Eğer $\psi_1(t), \psi_2(t), \dots \psi_n(t)$ işaretleri $[t_1, t_2]$ aralığında

$$\int_{t_1}^{t_2} \psi_m(t)\, \psi_n^*(t)\, dt = \begin{cases} 0 & m \neq n \\ E_m & m = n \end{cases}$$

sağlıyorsa ortagonaldir ilaveten
$\forall m \; E_m = 1$ ise işaretler otonormaldir

Örnek





$$\int_0^T x_1(t) \cdot x_2(t)\, dt$$

$$= \underbrace{\int_0^{T/2} A^2\, dt}_{A^2 \frac{T}{2}} + \underbrace{\int_{T/2}^{T} A(-A)\, dt}_{-A^2(T - T/2)}$$

$$= 0 \Rightarrow \text{Ortagonel}$$

$$E_1 = \int_0^T x_1^2(t)\, dt$$

$$= \int_0^T A^2\, dt = A^2 T = E_2$$

Ortonormal olması için iki işaretin genlikleri $\sqrt{A^2 T}$ ye bölmemiz lazım. Böylece enerjileri 1 olur.

$$x(t) \rightarrow E_x$$

$$x_N(t) = \frac{x(t)}{\sqrt{E_x}} \rightarrow E_{x_N} = \int_{-\infty}^{\infty} x_N^2(t)\,dt$$

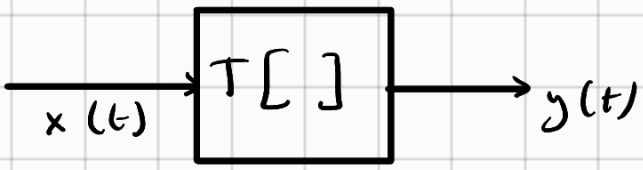
$$= \frac{1}{E_x} \underbrace{\int_{-\infty}^{\infty} x(t)^2\,dt}_{E_x}$$

$$= 1 \;//$$

Sistemler

Sistem, bir düzenektir.

$$y(t) = T[x(t)]$$



✓ Sürekli zamanlı s.

$$y(t) = \frac{d}{dt} x(t)$$

✗ Ayrık zamanlı s.

$$y[n] = x[n] - x[n-1]$$

✓ Lineer

✗ Lineer olmayan

α, β skaler

$$y_1(t) = T[x_1(t)]$$
$$y_2(t) = T[x_2(t)]$$

$$x_3(t) = \alpha x_1(t) + \beta x_2(t)$$

$$\rightarrow y_3(t) = T[x_3(t)]$$
$$= T[\alpha x_1(t) + \beta x_2(t)]$$
$$= \alpha T[x_1(t)] + \beta T[x_2(t)]$$
$$= \alpha y_1(t) + \beta y_2(t)$$

Eğer bir sistemin $x_1(t)$ ve $x_2(t)$ ye cevabını biliyorsanız sonsuz tane $x_3(t)$ x_1 ve x_2'nin lineer kombinasyonu cinsinden yazılabilir.

$y(t) = \frac{d}{dt} x(t)$ lineer

$y(t) = x^2(t)$ lineer değil

✗ Zamanla Değişen S. (Time variant) × Zamanla Değişmeyen S (Time Invariant)

$\forall t, x(t) \quad y(t) = T[x(t)]$

$T[x(t-t_0)] = y(t-t_0)$

$\Rightarrow$ Zamanla Değişmen (TI)

Örnek

$y(t) = t\,x(t)$

$x_1(t) \longrightarrow y_1(t) = t\,x_1(t)$

$x_2(t) = x(t-t_0) \longrightarrow y_2(t) = t\,x_2(t) = t\,x_1(t-t_0)$

$y_1(t-t_0) = (t-t_0)\,x_1(t-t_0) \neq$

$\Rightarrow$ Zamanla Değişen bir işaret

Ödev

$y(t) = x(t+1)$ Zamanla değişmeyen sistem

$x_1(t) \longrightarrow y_1(t) = x_1(t+1)$

$x_2(t) = x_1(t+1-t_0) \longrightarrow y_2(t) = x_2(t) = x_1(t+1-t_0)$

$y_1(t-t_0) = x_1(t+1-t_0)$

Zamanla değişmeyen sistem

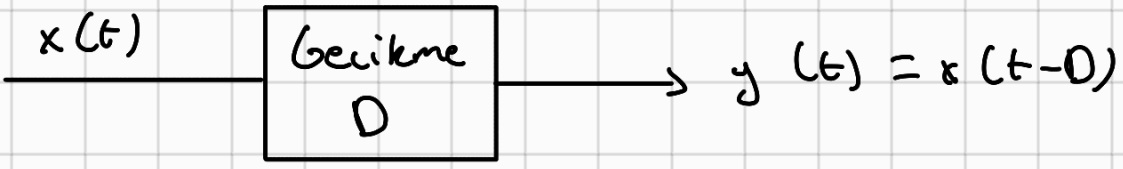
★ Lineer - Zamanla değişmen S = LZD sistem (Lineer Time Invariant (LTI))

Nedensel (Casual) Sistem

$y(t_0) = T[x(t), t \le t_0] \Rightarrow$ Nedensel sistem

Çıkışı sadece şimdiki ve geçmişteki ana bağlı çıkış olan sistemdir. Çıkış geleceğe bağlı değildir.

Örnek

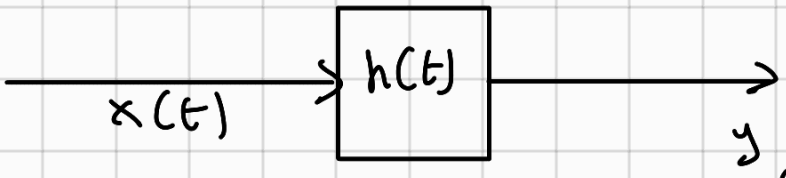


$D \ge 0$ için nedensel

Dürtü Cevabı (Impulse Response)

LZD sistem $\Rightarrow y(t) = T[\delta(t)] = h(t)$

$h(t)$: sistemin dürtü cevabı



$$y(t) = x(t) \circledast h(t)$$

$$= \int_{-\infty}^{\infty} x(z)\, h(t-z)\, dz$$

$$= \int_{-\infty}^{\infty} x(t-z)\, h(z)\, dz$$

LZD sistemin nedensel olması için gerek ve yeter şart $h(t) = 0,\ t < 0$

İspatı nedir??